EcoNest API Reference

Base URL: `http://localhost:3000`
Interactive docs (Swagger UI): `http://localhost:3000/api/docs`
Frontend: `http://localhost:5173` (Vite dev server, proxies `/api` to backend)

All responses follow the same envelope:

```
{ "success": true, "data": { ... } }
{ "success": false, "error": "message" }
```

Authentication

EcoNest uses **session tokens** — no JWTs or API keys. On login or register you receive a random token. Pass it in the `Authorization` header for protected endpoints:

```
Authorization: Bearer <token>
```

Tokens expire after **7 days**. Logging out immediately deletes the token from the database.

Auth Endpoints

POST /api/auth/register

Register a new user account.

Request body:

```
{
  "email": "user@example.com",
  "password": "SecurePass123!"
}
```

- `password` must be at least 8 characters
- Passwords are hashed with bcrypt (12 rounds) — the plain-text password is never stored

Response `201`:

```
{
  "success": true,
  "data": {
    "user": {
      "id": "uuid",
      "email": "user@example.com",
      "createdAt": "2026-03-13T10:00:00.000Z"
    },
    "token": "a3f9c2..."
  }
}
```

Errors:

Status	Meaning
`400`	Validation failed (invalid email / short password)
`409`	Email already registered

POST /api/auth/login

Log in with existing credentials.

Request body:

```
{
  "email": "user@example.com",
  "password": "SecurePass123!"
}
```

Response `200`:

```
{
  "success": true,
  "data": {
    "user": {
      "id": "uuid",
      "email": "user@example.com",
      "createdAt": "2026-03-13T10:00:00.000Z"
    },
    "token": "a3f9c2..."
  }
}
```

Errors:

Status	Meaning
`400`	Validation failed (invalid email / short password)
`401`	Invalid email or password

POST /api/auth/logout

🔒 Requires auth. Invalidates the current session token immediately.

Response `200`:

```
{ "success": true, "message": "Logged out successfully" }
```

Errors:

Status	Meaning
`401`	Missing, invalid, or expired token

GET /api/auth/me

🔒 Requires auth. Returns the currently authenticated user.

Response `200`:

```
{
  "success": true,
  "data": {
    "user": {
      "userId": "uuid",
      "email": "user@example.com"
    }
  }
}
```

Errors:

Status	Meaning
`401`	Missing, invalid, or expired token

Property Endpoints

For the current release, all property endpoints (`/trends` , `/heatmap` , `/properties` , `/regions` , `/stats`) are restricted to **2025 transaction data**.

GET /api/trends

Returns monthly average price trends. Optionally filtered.

Query parameters:

Parameter	Type	Description
`region`	string	City, district, or county name (case-insensitive)
`year`	string	Accepted value is `2025` (other values are ignored by the API)
`propertyType`	string	`D` Detached · `S` Semi-detached · `T` Terraced · `F` Flat · `O` Other

Response `200`:

```
{
  "success": true,
  "data": [
    {
```

```
      "year": 2025,
      "month": 1,
      "avgPrice": 285000,
      "transactionCount": 312,
      "priceGrowth": 1.4
    }
  ]
}
```

GET /api/heatmap

Returns average house price and total sales per county for the **2025 subset**. Useful for rendering a choropleth / heatmap.

Response `200`:

```
{
  "success": true,
  "data": [
    {
      "region": "GREATER LONDON",
      "avgPrice": 548000,
      "totalSales": 24800
    }
  ]
}
```

Results are ordered by `avgPrice` descending.

GET /api/properties

Paginated list of individual property sale records with optional filters. Results are restricted to **2025** transactions.

Query parameters:

Parameter	Type	Default	Description
`region`	string	—	City, district, or county
`minPrice`	integer	—	Minimum sale price (£)
`maxPrice`	integer	—	Maximum sale price (£)
`propertyType`	string	—	`D` / `S` / `T` / `F` / `O`
`page`	integer	`1`	Page number
`limit`	integer	`20`	Results per page (max 100)

Response `200`:

```
{
  "success": true,
  "data": {
    "items": [
      {
        "id": "uuid",
        "price": 325000,
        "transferDate": "2025-06-15T00:00:00.000Z",
        "postcode": "LS1 4AB",
        "propertyType": "T",
        "newBuild": false,
        "tenure": "Freehold",
        "street": "CHURCH ROAD",
        "townCity": "LEEDS",
        "district": "LEEDS",
        "county": "WEST YORKSHIRE"
      }
    ],
    "total": 18450,
    "page": 1,
    "limit": 20,
    "totalPages": 923
  }
}
```

GET /api/regions

Returns the list of all distinct counties present in the **2025 dataset subset**.

Response `200`:


```
{
  "success": true,
  "data": ["CAMBRIDGESHIRE", "CITY OF BRISTOL", "GREATER LONDON", "..."]
}
```

GET /api/stats

Returns top-level market statistics across the **2025 dataset subset**.

Response `200`:


```
{
  "success": true,
  "data": {
    "totalProperties": 482000,
    "avgPrice": 298500,
    "latestSaleDate": "2025-12-30T00:00:00.000Z"
```

```
  }
}
```

Market Insights

POST /api/insights/region

Returns a rich set of computed market insights for a given region, derived from the Land Registry data.

Request body:

```
{ "region": "LEEDS" }
```

The `region` field is matched case-insensitively against city name, district, and county.

Response `200`:

```
{
  "success": true,
  "data": {
    "region": "LEEDS",
    "totalTransactions": 18450,
    "avgPrice": 265000,
    "medianPrice": 240000,
    "minPrice": 42000,
    "maxPrice": 1850000,
    "priceRange": 1808000,
    "recentYearAvg": 278000,
    "oneYearGrowthPct": 3.2,
    "fiveYearGrowthPct": 18.7,
    "recentYearVolume": 2140,
    "affordabilityRatio": 7.9,
    "marketSignal": "moderate-growth",
    "newBuildShare": 12.4,
    "freeholdShare": 71.8,
    "propertyTypeBreakdown": [
      {
        "propertyType": "T",
        "label": "Terraced",
        "count": 6200,
        "avgPrice": 198000,
        "share": 33.6
      }
    ],
    "yearlyTrends": [
      {
        "year": 2024,
        "avgPrice": 270000,
        "transactionCount": 2250,
        "priceGrowthPct": 2.1
      }
```

```
    ]
  }
}
```

Field glossary:

Field	Description
`affordabilityRatio`	`avgPrice ÷ UK median household income (£35,000)`
`marketSignal`	Derived from 1-year growth: `strong-growth` (≥5%) · `moderate-growth` (≥1%) · `stable` (≥-1%) · `declining` · `insufficient-data`
`oneYearGrowthPct`	% change in avg price vs the prior 12-month period
`fiveYearGrowthPct`	% change comparing the most recent 12 months vs the period 5–6 years ago
`newBuildShare`	% of transactions that were new builds
`freeholdShare`	% of transactions with Freehold tenure

Errors:

Status	Meaning
`400`	`region` field missing from request body

Saved Listings

All saved listings endpoints require authentication.

POST /api/saved

🔒 Requires auth. Save a property to the current user's list.

Request body:

```
{
  "propertyId": "uuid",
  "notes": "Great location, near school"
}
```

- `propertyId` must be a valid UUID matching a record in the database
- `notes` is optional, max 500 characters

Response `201`:

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "userId": "uuid",
    "propertyId": "uuid",
```

```
    "notes": "Great location, near school",
    "createdAt": "2026-03-13T10:00:00.000Z",
    "property": { ... }
  }
}
```

Errors:

Status	Meaning
`400`	Validation failed
`401`	Not authenticated
`404`	Property ID not found
`409`	Property already saved

GET /api/saved

🔒 Requires auth. Returns all saved listings for the current user, newest first.

Response `200`:

```
{
  "success": true,
  "data": [
    {
      "id": "uuid",
      "notes": "Great location",
      "createdAt": "2026-03-13T10:00:00.000Z",
      "property": {
        "id": "uuid",
        "price": 325000,
        "street": "CHURCH ROAD",
        "townCity": "LEEDS",
        "county": "WEST YORKSHIRE"
      }
    }
  ]
}
```

PUT /api/saved/:id

🔒 Requires auth. Update the notes on a saved listing. Users can only update their own listings.

Request body:

```
{ "notes": "Updated note text" }
```

- `notes` can be a string (max 500 characters) or `null` to clear

Response `200`:

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "notes": "Updated note text",
    "createdAt": "2026-03-13T10:00:00.000Z",
    "property": { ... }
  }
}
```

Errors:

Status	Meaning
`400`	Validation failed
`401`	Not authenticated
`404`	Listing not found or belongs to another user

DELETE /api/saved/:id

🔒 Requires auth. Remove a saved listing by its ID. Users can only delete their own listings.

Response `200`:

```
{ "success": true, "message": "Listing removed" }
```

Errors:

Status	Meaning
`401`	Not authenticated
`404`	Listing not found or belongs to another user

Health Check

GET /api/health

No auth required. Confirms the server is running.

Response `200`:

```
{ "status": "ok", "timestamp": "2026-03-13T10:00:00.000Z" }
```

Frontend Pages

The Vue 3 frontend consumes all of the endpoints above and provides these pages:

Route	View	Description
`/login`	LoginView	Email + password sign in
`/register`	RegisterView	Account creation with confirmation
`/dashboard`	DashboardView	Stats cards, price trend chart, pie chart, volume chart, market insights assistant
`/trends`	TrendsView	Trend line chart with region / year / type filters and growth summary
`/heatmap`	HeatmapView	Horizontal bar chart by county with regional breakdown table
`/properties`	PropertiesView	Paginated property grid with save and compare buttons
`/saved`	SavedView	User's saved listings with notes and compare button
`/compare`	ComparisonView	Side-by-side comparison of up to 3 selected properties

Property Comparison (client-side feature)

The comparison tool is entirely client-side — it requires no additional API endpoints. Users select properties from the Properties or Saved pages. The comparison store (`stores/comparison.ts`) holds up to 3 `PropertySale` objects in memory. The ComparisonView renders a table with rows for price, type, location, tenure, new-build status, and sale date, plus insight cards showing the price range, best-value pick, and unique regions.